

A COMPLETE CHARACTERIZATION OF GEOMETRIC AUDIBILITY IN THE PLANE

RICK GULATI

ABSTRACT. A complete characterization of geometric audibility for bounded simply connected planar domains is established in three regimes. *Theorem A* (similarity rigidity): for every bounded planar domain that is star-shaped from some interior point, with piecewise $C^{2,\alpha}$ radial function bounded above and below, the Dirichlet spectrum determines the domain up to similarity. The proof factors through the inverse-square chord Laplacian Q on the boundary, a chord-arc identity $Q = \pi\Lambda + K$ with K bounded, the heat-trace bridge through a shared curvature-angle invariant ring $\mathcal{I}(\Gamma)$, cyclotomic rigidity at the regular attractor of each cyclic-symmetric stratum, and a golden-tetration contraction flow. A constructive reconstruction algorithm produces the domain explicitly from the spectrum, with end-to-end error $K^{-1/2}(\log K)^c$ in spectral truncation. *Theorem B* (sharpness): for every non-star-shaped bounded simply connected planar domain in a generic class, there exists a non-isometric isospectral partner, constructed explicitly via a permutation of boundary pieces that preserves $\mathcal{I}(\Gamma)$. The Gordon–Webb–Wolpert propellers [8, 1] are recovered as the canonical worked example. Star-shapedness is exactly the condition under which the polar parametrization forces uniqueness; a winding-number topological argument (analogous to the FTA proof) shows that the piece-multiplicity required for the partner construction is generic in the non-star-shaped class. *Theorem C* (Hessian rigidity): for every bounded piecewise smooth simply connected planar domain, the Dirichlet spectrum together with the Hadamard second variation $D^2\lambda_1$ of the first eigenvalue determines the domain up to isometry. The proof recovers the full boundary spectral data $\{\lambda_k, \partial_\nu u_k|_{\partial\Omega}\}$ from the Hadamard kernel via spectral subtraction: the boundary-restricted double-normal reduced resolvent is an order-one finite-part operator, a dressed chord Laplacian whose leading Laurent pole algebraically isolates $(\partial_\nu u_1)^2$. Belishev–Kurylev’s boundary control method then recovers the domain. Together, the three theorems form a complete characterization: the spectrum alone determines a domain up to similarity exactly in the star-shaped class; outside this class, additional variational data — specifically, the Hessian of λ_1 — is both sufficient and natural to break the Gordon–Webb–Wolpert obstruction.

CONTENTS

1. Introduction	3
1.1. The three regimes	3
1.2. Quantitative form and reconstruction algorithm	4
1.3. Topological characterization of audibility	4
1.4. The Q-operator framework	5
1.5. Relation to prior work	5
1.6. Organization	5
1.7. Notation and conventions	6

Date: May 12, 2026.

2020 Mathematics Subject Classification. Primary 35P05, 58J50; Secondary 11R29, 30C20, 35R30, 47A75, 57M50.

Key words and phrases. Inverse spectral geometry, Dirichlet eigenvalues, star-shaped domains, inverse-square Laplacian, Hadamard second variation, Sunada construction, Gordon–Webb–Wolpert, boundary control method, cyclotomic rigidity, Stark regulator, golden ratio.

2. The star-shaped class	6
2.1. Definition and basic structure	6
2.2. Stratified moduli	6
3. The unified Q operator	7
4. Chord-arc identity	7
4.1. Chord-arc bi-Lipschitz comparability	7
4.2. The identity	7
5. The Dirichlet-Q bridge	8
5.1. Heat traces	8
5.2. The shared invariant ring	8
5.3. The bridge theorem	9
6. Cyclotomic rigidity	9
6.1. Regular attractors	9
6.2. Rigidity at the attractor	9
7. The contraction flow	10
7.1. Universal scale-phase rigidity	10
7.2. Golden tetration base	10
7.3. The flow	10
7.4. Non-cyclic strata	11
8. Proof of Theorem A: similarity rigidity	11
9. The reconstruction algorithm	11
9.1. Overview	11
9.2. Stage A: Spectrum to heat-trace coefficients	11
9.3. Stage B: Heat coefficients to invariant ring	12
9.4. Stage C: Invariants to curvature and angles	12
9.5. Stage D: Frenet integration with corner jumps	12
9.6. Stage E: Verification via corner moments	12
9.7. End-to-end error	12
10. Proof of Theorem B: sharpness via Q-construction	12
10.1. Forward direction: Sunada-style construction	13
10.2. Reverse direction: star-shaped forces trivial π	13
10.3. Gordon-Webb-Wolpert worked example	13
11. Topological characterization: the winding-number argument	13
11.1. Radial projection and degree	13
11.2. The FTA-style count	14
11.3. Implication for piece-multiplicity	14
12. Proof of Theorem C: isometry rigidity via Hadamard data	14
12.1. Hadamard formula	14
12.2. Kernel decomposition	15
12.3. Boundary symbol: the dressed chord Laplacian	15
12.4. Spectral subtraction	15
12.5. Pointwise recovery of $\partial_\nu u_1$	15
12.6. Higher fluxes and boundary spectral data	16
12.7. Boundary control method	16
12.8. Proof of Theorem 1.3	16
12.9. Breaking the GWW degeneracy	17
13. The complete characterization	17
13.1. Connections to deeper mathematics	17
14. Open problems and obstructions	18

14.1. Non-cyclic strata	18
14.2. Conjecture 14.2	18
14.3. Cusps and reflex limits	18
14.4. Multiply connected domains	18
14.5. Higher dimensions	18
14.6. Numerical implementation of the Hadamard recovery	18
Appendix A. Finite-dimensional algebraic core	18
A.1. Directed inverse-square moments	19
A.2. The inverse-square operator	19
A.3. Bessel bridge and quantitative chart	20
Appendix B. Resolved reconstruction algorithm	20
B.1. Stage A: spectral zeta residues	20
B.2. Stage B: triangular inversion in the invariant ring	21
B.3. Stage C: curvature and angles	21
B.4. Stage D: Frenet integration	21
B.5. Stage E: corner-moment verification	22
B.6. Error balance	22
Appendix C. Sharpness and Sunada/Q reassembly	22
C.1. Heat-trace agreement	22
C.2. Boundary-piece permutations	22
C.3. Sunada symbol groups	22
C.4. Genericity scope	23
Appendix D. Hadamard boundary-control reconstruction	23
D.1. Moduli tangent spaces and the Hessian	23
D.2. Kernel decomposition	23
D.3. Boundary symbol and Laurent extraction	23
D.4. Recovering fluxes	24
D.5. From boundary spectral data to the domain	24
D.6. GWW degeneracy	24
References	24

1. INTRODUCTION

In 1966 Mark Kac [17] asked whether the Dirichlet spectrum of a bounded planar domain determines the domain up to isometry. Gordon, Webb, and Wolpert [8] answered negatively in 1992, constructing pairs of bounded simply connected planar domains with identical Dirichlet spectra and different shapes. The Buser–Conway–Doyle–Semmler simplification [1] reduces their construction to a pair of “propeller” polygons each composed of seven scalene triangles glued in different combinatorial patterns. The propellers share three central reflex (inward) corners: the central triangle’s three sides connect three inward corners of each propeller [1, §2].

The present paper establishes a complete characterization of geometric audibility for bounded simply connected planar domains, identifying both the maximal class on which the spectrum alone determines the shape and the minimal supplement that breaks the obstruction outside this class.

1.1. The three regimes.

Theorem 1.1 (Similarity rigidity on star-shaped domains). *Let $(\Omega_1, z_1), (\Omega_2, z_2)$ be bounded planar domains, each star-shaped from an interior point z_i , with radial function $r_i \in C_{\text{pw}}^{2,\alpha}(S^1)$*

satisfying $\|r_i\|_{C_{\text{pw}}^{2,\alpha}} + \|r_i^{-1}\|_{C^0} \leq R$. If $\text{Spec}(-\Delta_{\Omega_1}) = \text{Spec}(-\Delta_{\Omega_2})$, then $\Omega_2 = g \cdot \Omega_1$ for some $g \in \text{Sim}(\mathbb{R}^2)$.

Theorem 1.2 (Sharpness via Q-construction). *For a generic bounded piecewise smooth simply connected planar domain Ω that is not star-shaped from any interior point, there exists a non-isometric isospectral partner Ω' , constructed explicitly as a permutation of the boundary pieces of $\partial\Omega$ preserving the curvature-angle invariant ring $\mathcal{I}(\Gamma)$ together with area and perimeter. The Gordon–Webb–Wolpert propellers are the canonical worked example.*

Theorem 1.3 (Isometry rigidity via Hadamard data). *Let $\Omega_1, \Omega_2 \subset \mathbb{R}^2$ be bounded piecewise $C^{2,\alpha}$ simply connected planar domains with interior angles at corners in $(0, 2\pi) \setminus \{\pi\}$. If*

(i) $\text{Spec}(-\Delta_{\Omega_1}) = \text{Spec}(-\Delta_{\Omega_2})$, and

(ii) $D^2\lambda_1|_{\Omega_1}$ and $D^2\lambda_1|_{\Omega_2}$ are isomorphic as quadratic forms on the moduli tangent spaces,

then $\Omega_2 = g \cdot \Omega_1$ for some $g \in \text{Iso}(\mathbb{R}^2)$.

Theorems 1.1–1.3 together form a trichotomy: the spectrum alone is sufficient for shape determination up to similarity *exactly* on the star-shaped class (Theorem 1.1, with sharpness from Theorem 1.2); outside this class, the Gordon–Webb–Wolpert obstruction is broken by the addition of the Hadamard second variation $D^2\lambda_1$, giving isometry rigidity on the full piecewise smooth class (Theorem 1.3).

1.2. Quantitative form and reconstruction algorithm. Theorem 1.1 is constructive: a finite algorithm $\mathcal{A}$ takes the truncated Dirichlet spectrum $(\lambda_1, \dots, \lambda_K)$ as input and outputs $\widehat{\Omega} \in \mathcal{S}_R$ with

$$d_H(\partial\widehat{\Omega}, \partial\Omega) \leq C_R K^{-1/2} (\log K)^c$$

in Hausdorff distance modulo similarity, for an explicit constant c depending on the analytic radius of the boundary curvature. The algorithm proceeds in five stages: Mellin extraction of Berry heat-trace coefficients (Section 9.2), triangular inversion to integrated invariants (Section 9.3), Edward–Polterovich–Sher Fourier inversion for curvature and Newton-symmetric inversion for corner angles (Section 9.4), piecewise Frenet integration with Gauss–Bonnet closure (Section 9.5), and verification through corner inverse-square moments (Section 9.6).

1.3. Topological characterization of audibility. The star-shaped hypothesis admits a topological characterization. For an interior point $z_0 \in \Omega$ and the radial projection $\pi_{z_0}: \partial\Omega \rightarrow S^1$, $\pi_{z_0}(w) = \arg(w - z_0)$, Ω is star-shaped from z_0 if and only if π_{z_0} is a diffeomorphism. The map has degree 1 (the winding number of $\partial\Omega$ around z_0). For non-star-shaped Ω , π_{z_0} has critical points where the radial line is tangent to $\partial\Omega$.

Theorem 1.4 (Topological piece-multiplicity). *For every bounded piecewise smooth simply connected planar domain Ω that is not star-shaped from any interior point, and for every $z_0 \in \Omega$, the radial projection π_{z_0} has at least 4 critical points. Equivalently, $\partial\Omega$ admits a decomposition into at least 4 pieces that are locally star-shaped from z_0 .*

This is the topological analog of the fundamental theorem of algebra: a polynomial of degree $n \geq 1$ has at least n zeros (counted with multiplicity) because the winding number of the image of a large circle is n . Here, the radial projection's degree 1 combined with non-trivial fold structure forces ≥ 4 critical points via a signed-index argument.

Theorem 1.4 provides the topological foundation for the genericity claim in Theorem 1.2: the piece-multiplicity required for the construction of an isospectral partner is forced by the failure of star-shapedness.

1.4. The Q -operator framework. The proofs of all three main theorems factor through a common object: the inverse-square chord Laplacian Q on the boundary. For a bounded planar domain with piecewise smooth boundary,

$$Q_\Gamma = Q_\Gamma^{\text{arc}} + Q_\Gamma^{\text{corner}} + Q_\Gamma^{\text{mix}},$$

combining a hypersingular boundary integral on smooth arcs, an inverse-square vertex-pair matrix at corners, and a cross-coupling. The chord-arc identity

$$Q_\Gamma = \pi\Lambda_\Gamma + K_\Gamma$$

with K_Γ bounded on $L^2(\Gamma)$ (uniformly in the class $\mathcal{S}_R$) connects Q to the Steklov Dirichlet-to-Neumann operator Λ_Γ . The Dirichlet spectrum determines the Q heat trace through a shared ring of integrated curvature-angle invariants $\mathcal{I}(\Gamma)$. From the Q spectrum, the boundary curvature and corner angles are recovered, and the boundary is reconstructed by piecewise Frenet integration.

The same operator Q provides the structural mechanism for both the positive direction (Theorem 1.1) and the sharpness direction (Theorem 1.2): two boundaries with the same $\mathcal{I}(\Gamma)$ have the same Q heat trace and hence the same Dirichlet spectrum; for star-shaped boundaries the polar parametrization forces uniqueness, while for non-star-shaped boundaries the piece-multiplicity allows multiple reassemblies of the same $\mathcal{I}$ -data.

The third theorem (Theorem 1.3) goes beyond the heat-trace Q framework but uses the same leading operator. The Hadamard quadratic form $D^2\lambda_1$ encodes the reduced boundary resolvent

$$G_R = \partial_\nu(-\Delta - \lambda_1)^{-1}_{\perp u_1} \partial_\nu,$$

whose boundary kernel is not logarithmic: it is a pseudodifferential kernel of order $+1$ with the same hypersingular principal part as the chord Laplacian. Thus $D^2\lambda_1$ is a flux-dressed Q operator, and its leading Laurent pole recovers $\partial_\nu u_1$ before the remaining spectral decomposition recovers all higher boundary fluxes.

1.5. Relation to prior work. The Kac question for convex polygons and smooth strictly convex curves has been considered in restricted classes. The general problem on the convex class remains open in the literature; see [21, §1] and the survey [7]. The companion papers [10, 11, 12] address subclasses (convex polygons, smooth star-shaped curves, piecewise smooth convex). The present paper unifies these results under the single star-shaped hypothesis and complements them with the converse (Theorem 1.2) and the full-class reconstruction via Hadamard data (Theorem 1.3).

For the inverse boundary spectral problem on bounded domains with prescribed boundary data, Belishev's boundary control method [2] and Sylvester–Uhlmann's solution of the Calderón problem [30] establish that the Dirichlet-to-Neumann map determines the domain up to isometry. Theorem 1.3 reduces the inverse spectral problem on bounded domains to this boundary-control problem by extracting the boundary spectral data from the spectrum-plus-Hessian.

1.6. Organization. Section 2 defines the star-shaped class $\mathcal{S}_R$ and its stratified moduli space. Section 3 defines the unified Q operator. Section 4 establishes the chord-arc identity. Section 5 establishes the Dirichlet-to- Q bridge through the invariant ring $\mathcal{I}(\Gamma)$. Section 6 proves cyclo-tomic rigidity at regular attractors. Section 7 constructs the contraction flow. Section 8 proves Theorem 1.1. Section 9 gives the constructive reconstruction algorithm with stages A–E. Section 10 proves Theorem 1.2, including the Gordon–Webb–Wolpert worked example. Section 11 proves Theorem 1.4 via winding numbers. Section 12 proves Theorem 1.3 via Hadamard kernel decomposition and boundary control. Section 13 summarizes the complete characterization. Section 14 records open problems. The appendices collect the companion-paper details: the

finite-dimensional algebraic core, the resolved reconstruction algorithm, the Sunada/ $\mathbb{Q}$ sharpness construction, and the Hadamard boundary-control reconstruction.

1.7. Notation and conventions. Throughout, $\varphi = (1 + \sqrt{5})/2$ denotes the golden ratio, $\mu_\star = 1/(2\varphi)$, $b_\star = \exp(\mu_\star e^{-\mu_\star})$ the golden tetration base, and η_n the fundamental unit of $\mathbb{Q}(\zeta_n)^+$ (real cyclotomic subfield). Arc-length on a piecewise smooth curve is s , with $L = |\Gamma|$. The Fourier basis on $S_L^1 = \mathbb{R}/L\mathbb{R}$ is $e_k(s) = e^{2\pi i k s/L}$. For a planar domain Ω , $\lambda_k(\Omega)$ denotes the k -th Dirichlet eigenvalue and u_k the corresponding L^2 -normalized eigenfunction.

2. THE STAR-SHAPED CLASS

2.1. Definition and basic structure.

Definition 2.1. A pair (Ω, z_0) with $\Omega \subset \mathbb{R}^2$ bounded and open and $z_0 \in \Omega$ belongs to $\mathcal{S}_R$ if:

- (i) $\partial\Omega - z_0 = \{r(\theta)e^{i\theta} : \theta \in S^1\}$ for some function $r : S^1 \rightarrow (0, \infty)$.
- (ii) r is piecewise $C^{2,\alpha}$: finitely many polar corners $\theta_1 < \dots < \theta_n$ with r in $C^{2,\alpha}$ on each open arc (θ_l, θ_{l+1}) and finite one-sided limits of r, r' at each θ_l .
- (iii) $\sup_\theta r(\theta) + \sup_\theta r(\theta)^{-1} + \sup_{\text{arcs}} \|r\|_{C^{2,\alpha}} \leq R$.

The points $v_l := r(\theta_l)e^{i\theta_l} + z_0$ are *Cartesian corners* of $\partial\Omega$, with interior angles $\alpha_l \in (0, 2\pi) \setminus \{\pi\}$. The smooth arcs $\gamma_l := \{r(\theta)e^{i\theta} + z_0 : \theta \in (\theta_l, \theta_{l+1})\}$ join consecutive corners.

Proposition 2.2. For $(\Omega, z_0) \in \mathcal{S}_R$:

- (i) $\partial\Omega$ is a piecewise $C^{2,\alpha}$ Jordan curve, and Ω is bi-Lipschitz to the unit disk with constants depending only on R .
- (ii) The interior angles $\alpha_l \in (0, 2\pi) \setminus \{\pi\}$; convex corners have $\alpha_l \in (0, \pi)$, reflex corners have $\alpha_l \in (\pi, 2\pi)$.
- (iii) Ω may be non-convex (star polygons, smooth star-shaped non-convex domains are in $\mathcal{S}_R$).

2.2. Stratified moduli.

Definition 2.3. A *stratum type* J records: the number n of polar corners; the cyclic-and-reflection equivalence class of polar-corner positions; the convex/reflex type of each Cartesian corner; the analytic regularity of each arc. The stratum $\mathcal{M}_{n,J} \subset \mathcal{S}_R/\text{Sim}$ collects similarity classes of domains with combinatorial type J .

Proposition 2.4. Each $\mathcal{M}_{n,J}$ is a real-analytic manifold (smooth Fréchet manifold when arc shape is infinite-dimensional). The closure of $\mathcal{M}_{n,J}$ contains adjacent strata in which corners merge or split.

The recovered classes:

Smooth star-shaped ($n = 0$): one closed smooth arc, no corners. Includes the disk, smooth strictly convex curves, smooth non-convex stars.

Convex polygons: r piecewise linear, all corners convex ($\alpha_l \in (0, \pi)$).

Star polygons: r piecewise linear, alternating convex and reflex corners (e.g., regular pentagram has interior angles $\pi/5$ and $7\pi/5$).

Piecewise smooth convex: smooth arcs with $\alpha_l \in (0, \pi)$. Includes hearts, lenses, stadiums with corners.

Piecewise smooth non-convex star-shaped: mixed arcs with some reflex angles, but star-shaped from a center.

3. THE UNIFIED Q OPERATOR

Definition 3.1. For $(\Omega, z_0) \in \mathcal{S}_R$ with Cartesian corners $v_1, \dots, v_n$ and smooth arcs $\gamma_1, \dots, \gamma_n$, the *inverse-square chord Laplacian* on $L^2(\Gamma)$ is the symmetric densely-defined operator

$$Q_\Gamma f = Q_\Gamma^{\text{arc}} f + Q_\Gamma^{\text{corner}} f + Q_\Gamma^{\text{mix}} f,$$

where:

$$(Q_\Gamma^{\text{arc}} f)(x) = \text{p.v.} \int_\Gamma \frac{f(x) - f(y)}{|x - y|^2} d\sigma(y) \quad (x \in \text{smooth part}),$$

$$(Q_\Gamma^{\text{corner}})_{ll'} = \begin{cases} -|v_l - v_{l'}|^{-2}, & l \neq l', \\ \sum_{l'' \neq l} |v_l - v_{l''}|^{-2}, & l = l', \end{cases}$$

and Q_Γ^{mix} provides the cross-coupling between arc-trace and vertex-trace data.

Proposition 3.2. Q_Γ is symmetric on its natural domain, $Q_\Gamma \mathbf{1} = 0$, $Q_\Gamma \succeq 0$ on $\mathbf{1}^\perp$, with kernel $\mathbb{R} \cdot \mathbf{1}$. Its spectrum $\{\sigma_k(Q_\Gamma)\}$ is discrete with $\sigma_k \rightarrow \infty$.

Proof. Symmetry: by definition. Kernel: $Q_\Gamma \mathbf{1} = 0$ from row-sum cancellation. Positivity on $\mathbf{1}^\perp$: the energy form

$$\langle f, Q_\Gamma f \rangle = \frac{1}{2} \int \int_{\Gamma \times \Gamma} \frac{(f(x) - f(y))^2}{|x - y|^2} d\sigma(x) d\sigma(y)$$

is manifestly non-negative, with equality iff f is constant.

Discrete spectrum: Q_Γ is a positive self-adjoint pseudodifferential operator of order 1 on a compact 1-manifold (the principal symbol is $\pi|\xi|$, see Section 4); standard elliptic theory gives discrete spectrum. $\square$

Remark 3.3 (Specialisations). On a smooth star-shaped boundary ($n = 0$): $Q_\Gamma = Q_\Gamma^{\text{arc}}$, the hypersingular boundary integral operator. On a convex polygon ($\kappa \equiv 0$ on arcs, all arcs are line segments): Q_Γ restricted to the vertex-trace data is the matrix Q_Γ^{corner} of [10].

4. CHORD-ARC IDENTITY

4.1. Chord-arc bi-Lipschitz comparability.

Lemma 4.1. For $(\Omega, z_0) \in \mathcal{S}_R$, there exist $c_R, C_R > 0$ such that for every $x, y \in \Gamma$,

$$c_R d_\Gamma(x, y) \leq |x - y| \leq C_R d_\Gamma(x, y),$$

where d_Γ is the intrinsic arc-length distance.

Proof. The upper bound is by the triangle inequality. The lower bound uses $\inf r \geq R^{-1}$ together with the bounded radial logarithmic derivative; the constant c_R depends only on R . Strict convexity is not required: the bound is a property of the radial parametrization, not of geometric convexity. $\square$

4.2. The identity.

Theorem 4.2. For every $(\Omega, z_0) \in \mathcal{S}_R$, $Q_\Gamma = \pi \Lambda_\Gamma + K_\Gamma$, where Λ_Γ is the Steklov Dirichlet-to-Neumann operator on Γ and K_Γ is bounded on $L^2(\Gamma)$ with operator norm bounded by a constant depending only on R . The symbol of K_Γ is:

- (i) On smooth arcs: $\langle K_\Gamma e_k, e_k \rangle = \pi \kappa(s)^2 / 12 + O(|k|^{-2})$ in the arc-length Fourier frame.
- (ii) At a corner of interior angle $\alpha \in (0, 2\pi) \setminus \{\pi\}$: K_Γ has Costabel–Stephan Mellin symbol $M_\alpha^K(\zeta) = \pi / \sin(\pi \zeta) - \pi M_\alpha^\Lambda(\zeta)$, bounded on the critical line $\Re \zeta = 1/2$.

Consequently $\sigma_k(Q_\Gamma) = \pi \sigma_k(\Lambda_\Gamma) + O(1)$ as $k \rightarrow \infty$.

Proof. On smooth arcs: the Frenet expansion $z(s) - z(s') = -uT(s) + \frac{1}{2}i\kappa u^2 T(s) + O(u^3)$ with $u = s - s'$ gives the u^{-2} singularity matching $\pi|\xi|$ after Hadamard regularization (cf. [24, Theorem 8.12]). The subprincipal $\pi\kappa^2/12$ comes from the u^0 coefficient of the kernel expansion.

At corners: the Mellin calculus of Costabel–Stephan [5] gives the wedge symbol. For convex corners ($\alpha \in (0, \pi)$) the symbol is bounded on the critical strip; the same calculus extends to reflex corners ($\alpha \in (\pi, 2\pi)$) by direct computation of the wedge harmonic continuation. The boundedness of $M_\alpha^K(\zeta)$ on $\Re \zeta = 1/2$ uses the pole-cancellation between $\pi/\sin(\pi\zeta)$ and the Steklov Mellin symbol.

The chord-arc inequality Lemma 4.1 bounds K_Γ uniformly on $\mathcal{S}_R$. $\square$

5. THE DIRICHLET-Q BRIDGE

5.1. Heat traces. For $(\Omega, z_0) \in \mathcal{S}_R$, three heat traces are relevant:

- Dirichlet: $Z_\Omega^{\text{Dir}}(t) = \sum_{k \geq 1} e^{-\lambda_k(\Omega)t}$.
- Steklov: $Z_\Gamma^\Lambda(t) = \sum_{k \geq 0} e^{-\sigma_k(\Lambda_\Gamma)t}$.
- Q-operator: $Z_\Gamma^Q(t) = \sum_{k \geq 0} e^{-\sigma_k(Q_\Gamma)t}$.

Proposition 5.1 (Berry expansion).

$$Z_\Omega^{\text{Dir}}(t) \sim \frac{|\Omega|}{4\pi t} - \frac{|\Gamma|}{8\sqrt{\pi t}} + \frac{1}{12\pi} \int_\Gamma^{\text{smooth}} \kappa ds + \sum_l \frac{\pi^2 - \alpha_l^2}{24\pi\alpha_l} + \sum_{j \geq 1} b_j^{\text{Dir}}(\Gamma) t^{j/2}.$$

The wedge contribution $(\pi^2 - \alpha^2)/(24\pi\alpha)$ extends to $\alpha \in (0, 2\pi)$ — positive for convex corners, negative for reflex corners — consistent with the Carslaw–Jaeger wedge heat kernel [3].

Proof. McKean–Singer / Berry expansion (cf. [25, §4.2]). The convex-corner case was made precise by van den Berg–Srisatkunaratjah [31]; the reflex case follows by the same wedge heat kernel argument extended to $\alpha > \pi$, with the trigonometric formula extending continuously past $\alpha = \pi$. $\square$

Proposition 5.2 (Steklov heat trace). *For Γ real-analytic in each arc and corners in $(0, 2\pi) \setminus \{\pi\}$,*

$$Z_\Gamma^\Lambda(t) \sim \frac{|\Gamma|}{\pi t} + \sum_l c_{\text{corner}}^\Lambda(\alpha_l) + \sum_{j \geq 0} a_j(\Gamma) t^j,$$

with $a_j(\Gamma) = \int_\Gamma^{\text{smooth}} P_j^{\text{Stek}}(\kappa, \kappa', \dots) ds + (\text{corner trig.})$ universal polynomials [6, 27, 22].

Proposition 5.3 (Q heat trace).

$$Z_\Gamma^Q(t) = Z_\Gamma^\Lambda(\pi t) + R_K(t),$$

where $R_K(t) \sim \sum_l r_{\text{corner}}(\alpha_l) + \sum_{j \geq 0} k_j(\Gamma) t^j$ from the Duhamel expansion of K_Γ .

5.2. The shared invariant ring.

Definition 5.4. The *curvature-angle invariant ring* of Γ is

$$\mathcal{I}(\Gamma) := \mathcal{I}_{\text{smooth}}(\Gamma) \oplus \mathcal{I}_{\text{corner}}(\Gamma),$$

where:

$$\begin{aligned} \mathcal{I}_{\text{smooth}}(\Gamma) &:= \mathbb{R} \left[\sum_l \int_{\gamma_l} \kappa^{i_0}(\kappa')^{i_1} \dots ds \right] / \text{IBP}, \\ \mathcal{I}_{\text{corner}}(\Gamma) &:= \mathbb{R} \left[\sum_l p(\alpha_l) : p \in \mathbb{R}[\sin, \cos, \cot, \csc] \right] / \text{trig identities}, \end{aligned}$$

augmented by $L = |\Gamma|$.

Theorem 5.5. *For every j , each of $b_j^{\text{Dir}}(\Gamma)$, $a_j(\Gamma)$, $c_j^Q(\Gamma) = \pi^j a_j + k_j$ lies in $\mathcal{I}(\Gamma)$ as a universal polynomial expression.*

Proof. Each heat trace is constructed from local parametrices: smooth-arc parametrix depending only on κ and arc-length derivatives, and corner-wedge parametrix depending only on α . Trace against the diagonal produces integrated polynomial-in-curvature and trigonometric-in-angle expressions, all in $\mathcal{I}(\Gamma)$. $\square$

5.3. The bridge theorem.

Theorem 5.6. *For $\Omega \in \mathcal{S}_R$ with real-analytic smooth arcs, $\text{Spec}(-\Delta_\Omega)$ determines $\text{Spec}(Q_\Gamma)$ as a multiset.*

Proof. $\text{Spec}(-\Delta_\Omega)$ determines $Z_\Omega^{\text{Dir}}(t)$, hence the Berry coefficients $\{b_j^{\text{Dir}}\}$ via Mellin residues $b_j^{\text{Dir}} = \text{Res}_{s=-j/2} \Gamma(s) \zeta_\Omega^{\text{Dir}}(s)$.

The triangular structure of Theorem 5.5 together with Fourier inversion on each real-analytic arc (Edward / Polterovich–Sher) and Newton-symmetric inversion on corner angles recovers $\mathcal{I}(\Gamma)$ in its entirety: smooth-arc curvature κ_l on each arc, corner angles $\{\alpha_l\}$, perimeter L .

$\mathcal{I}(\Gamma)$ then determines every coefficient c_j^Q in the Q heat-trace expansion (Theorem 5.5). The Q heat trace $Z_\Gamma^Q(t)$ is recovered by Borel summability in the real-analytic class, and as the Laplace transform of the spectral counting measure, it uniquely determines $\text{Spec}(Q_\Gamma)$. $\square$

6. CYCLOTOMIC RIGIDITY

6.1. Regular attractors. For a cyclic-symmetric stratum $\mathcal{M}_{n,J} \subset \mathcal{S}_R/\text{Sim}$ with maximal cyclic-and-reflection group $D_{n'}$, the *regular attractor* $\Omega_{n,J}^*$ is the unique $D_{n'}$ -symmetric domain in $\mathcal{M}_{n,J}$:

- Regular convex n -gon for the all-corner stratum with all $\alpha_l = (n-2)\pi/n$.
- Disk for the smooth no-corner stratum.
- Regular star polygon $\{n/k\}$ for the cyclic-symmetric polygonal stratum with mixed corner types.
- Regular rounded polygon (cyclic-symmetric mixed arc-and-corner) for the cyclic-symmetric piecewise smooth stratum.

6.2. Rigidity at the attractor.

Theorem 6.1. *At the regular attractor $\Omega_{n,J}^*$ with cyclic symmetry $D_{n'}$, the differential $DQ|_{\Omega_{n,J}^*}$ on the similarity-gauged tangent space $T_{\Omega_{n,J}^*} \mathcal{M}_{n,J}$ is injective, with injectivity modulus bounded below by $\eta_{n'}^{-1}$, where $\eta_{n'}$ is the fundamental unit of $\mathbb{Q}(\zeta_{n'})^+$. In particular, $\eta_5 = \varphi$ gives the rate $\varphi^{-1} \approx 0.618$ at the regular pentagon.*

Proof. Mode decomposition. The kernel condition for DQ at the regular attractor decouples in the discrete-and-continuous Fourier basis on Γ under the $D_{n'}$ symmetry. Both vertex and arc Fourier modes are eigendirections of the DQ -form, each killed independently.

Vertex part. The kernel condition reduces to $\Re[a_k(z_i + z_j)] = 0$ for chord pairs (z_i, z_j) at the attractor vertices (the n -th roots of unity for the regular polygon, or appropriate cyclotomic positions for star polygons). The chord-pair magnitudes lie in $\mathbb{Q}(\zeta_{n'})^+$, with the smallest non-trivial magnitude $\geq \eta_{n'}^{-1}$. Galois conjugacy of distinct chord-pair magnitudes over $\mathbb{Q}$ supplies linear independence:

For $n' = 5$: chord-pair magnitudes are φ and φ^{-1} , Galois conjugates with trace $\varphi + \varphi^{-1} = \sqrt{5}$ and norm $\varphi \cdot \varphi^{-1} = 1$. The trace identity $\varphi^2 + \varphi^{-2} = 3$ implies φ, φ^{-1} are $\mathbb{Q}$ -linearly independent. The orbits $\{z_i + z_j\}$ at fixed chord-pair magnitude span $\mathbb{R}^2$ over $\mathbb{R}$, forcing $a_k = 0$.

Arc part. The Bessel-bridge isotype multipliers κ_m^{disk} at the disk [11, §5] provide the lower bound $\kappa_m^{\text{disk}} \geq 68.7$ uniformly. For mixed strata, the arc-mode contribution to the injectivity modulus is bounded uniformly below.

Mixing part. The cross-coupling between vertex and arc modes (via Q_{Γ}^{mix}) preserves the kernel decoupling, by character orthogonality on the $D_{n'}$ representation.

Conclusion. Total injectivity modulus $\geq \min(\eta_{n'}^{-1}, c\kappa_m^{\text{disk}}/m) = \eta_{n'}^{-1}$ for $n' \geq 5$. $\square$

Remark 6.2. At the regular star polygon $\{n/k\}$ with reflex inner corners, the vertices remain at n -th roots of unity (with cyclic reordering). The Galois pairing argument is identical; reflex angles do not weaken the rigidity rate.

7. THE CONTRACTION FLOW

7.1. Universal scale-phase rigidity.

Theorem 7.1 ([14, Theorem 9.15]). *Let $T: U \rightarrow \mathbb{C}$ be holomorphic with hyperbolic fixed point z^* of multiplier λ , $|\lambda| \notin \{0, 1\}$. There exists a unique holomorphic family of fractional iterates $\{T^{[h]}\}$ on the linearization domain, satisfying the group law $T^{[h_1+h_2]} = T^{[h_1]} \circ T^{[h_2]}$, joint holomorphy in (h, z) , $T^{[0]} = \text{id}$, $T^{[1]} = T$, and $(T^{[h]})'(z^*) = \lambda^h$ on the principal branch.*

7.2. Golden tetraton base.

Theorem 7.2 ([14, Theorem 9.38]). *For the multiplier $\mu_\star = 1/(2\varphi)$, there exists a unique base $b_\star \in (e^{-e}, e^{1/e})$ such that the operator-canonical fractional iterate of $z \mapsto b_\star^z$ at its principal hyperbolic fixed point has multiplier $\mu_\star$. The closed form is*

$$b_\star = \exp(\mu_\star e^{-\mu_\star}) \approx 1.25467, \quad z^\dagger(b_\star) = e^{1/(2\varphi)}.$$

7.3. The flow.

Definition 7.3. For each cyclic-symmetric stratum $\mathcal{M}_{n,J} \subset \mathcal{S}_R/\text{Sim}$ with symmetry $D_{n'}$ and regular attractor $\Omega_{n,J}^\star$, the *stratified contraction flow* is

$$T_h^{(n,J)}: \mathcal{M}_{n,J} \rightarrow \mathcal{M}_{n,J}, \quad T_h^{(n,J)}(\Omega) := (\sigma^{(n,J)})^{-1}((\mu_\star^{(n,J)})^h \sigma^{(n,J)}(\Omega)),$$

the operator-canonical fractional-iterate family of Theorem 7.1 at multiplier $\mu_\star^{(n,J)} := 1/(2\eta_{n'})$, transported through the shape-Hessian gauge $\sigma^{(n,J)}$ from Theorem 6.1.

Theorem 7.4. $T_h^{(n,J)}$ satisfies:

- (i) $T_0^{(n,J)} = \text{id}$, $T_h^{(n,J)}(\Omega_{n,J}^\star) = \Omega_{n,J}^\star$, $T_h^{(n,J)}(\Omega) \rightarrow \Omega_{n,J}^\star$ as $h \rightarrow \infty$ at rate $(2\eta_{n'})^{-h}$.
- (ii) $T_h^{(n,J)}(\mathcal{M}_{n,J}) \subseteq \mathcal{M}_{n,J}$.
- (iii) $h \mapsto T_h^{(n,J)}$ real-analytic; $T_h^{(n,J)}$ real-analytic on $\mathcal{M}_{n,J}$.
- (iv) Group law: $T_{h_1+h_2}^{(n,J)} = T_{h_1}^{(n,J)} \circ T_{h_2}^{(n,J)}$.

Remark 7.5. $-\log \mu_\star^{(n,J)} = \log(2\eta_{n'}) = \log 2 + L'(0, \chi_{n'})$, the Stark regulator of $\mathbb{Q}(\zeta_{n'})^+$ [28]. At $n' = 5$: $\log \varphi = L'(0, \chi_5)$, the Stark regulator of $\mathbb{Q}(\sqrt{5})$ [4].

7.4. Non-cyclic strata.

Conjecture. On every stratum $\mathcal{M}_{n,J}$, including non-cyclic-symmetric ones, there exists a real-analytic contraction flow $T_h^{(n,J)}$ converging to the stratum's closure point on its higher-symmetry boundary.

The cyclic-symmetric strata (containing all regular and dihedrally-symmetric shapes) are handled unconditionally by Theorem 7.4. The non-cyclic strata require this conjecture for the unconditional global statement of Theorem 1.1.

8. PROOF OF THEOREM A: SIMILARITY RIGIDITY

Fix a cyclic-symmetric stratum $\mathcal{M}_{n,J}$ and the exceptional locus

$$Z_{n,J} := \{\Omega \in \mathcal{M}_{n,J} : \text{Spec}(-\Delta_\Omega) \text{ fails to determine } \Omega \bmod \text{Sim}(\mathbb{R}^2)\}.$$

Real-analyticity of $Z_{n,J}$. The eigenvalue map is real-analytic on each stratum by Rellich–Kondrat’ev analyticity [18]. The exceptional condition is the vanishing of a real-analytic Jacobian determinant; $Z_{n,J}$ is closed and real-analytic.

$T_h^{(n,J)}$ -invariance. The data map factors as

$$\text{Spec}(-\Delta_\Omega) \xrightarrow{\text{Thm. 5.6}} \text{Spec}(Q_\Gamma) \xrightarrow{\text{Thm. 4.2}} \text{Steklov heat trace} \rightarrow (\kappa, \{\alpha_l\}) \xrightarrow{\text{FTC2}} \Gamma,$$

all real-analytic. The flow $T_h^{(n,J)}$ acts in σ -coordinates by multiplicative scaling, preserving real-analytic loci. Hence $T_h^{(n,J)}(Z_{n,J}) \subseteq Z_{n,J}$.

Attractor in closure. If $Z_{n,J} \neq \emptyset$, then for $\Omega \in Z_{n,J}$, $T_h^{(n,J)}(\Omega) \rightarrow \Omega_{n,J}^*$ (Theorem 7.4(i)), and $Z_{n,J}$ being closed forces $\Omega_{n,J}^* \in Z_{n,J}$.

Attractor not exceptional. By Theorem 6.1, $DQ|_{\Omega_{n,J}^*}$ has injectivity modulus $\eta_{n'}^{-1} > 0$; the data map is locally invertible at $\Omega_{n,J}^*$, so $\Omega_{n,J}^* \notin Z_{n,J}$.

The contradiction forces $Z_{n,J} = \emptyset$ on every cyclic-symmetric stratum. Theorem 1.1 holds unconditionally on cyclic-symmetric strata and conditionally on the contraction-flow conjecture on the remaining strata. $\square$

9. THE RECONSTRUCTION ALGORITHM

We give the constructive content of Theorem 1.1: an explicit algorithm taking $\text{Spec}(-\Delta_\Omega)$ as input and producing Ω as output, with quantitative error bounds.

9.1. Overview. The algorithm proceeds in five stages:

$$\text{Spec}(-\Delta_\Omega) \xrightarrow{\text{A}} \{b_j^{\text{Dir}}\} \xrightarrow{\text{B}} \mathcal{I}(\Gamma) \xrightarrow{\text{C}} (\kappa, \{\alpha_l\}) \xrightarrow{\text{D}} \Gamma \xrightarrow{\text{E}} \text{verification}.$$

9.2. Stage A: Spectrum to heat-trace coefficients.

Lemma 9.1. For $j \geq 0$, the Berry coefficient $b_j^{\text{Dir}}(\Gamma)$ is recovered from the spectrum via

$$b_j^{\text{Dir}}(\Gamma) = \text{Res}_{s=-j/2} \Gamma(s) \zeta_\Omega^{\text{Dir}}(s).$$

For truncated spectrum $(\lambda_1, \dots, \lambda_K)$, the estimate $\widehat{b}_j^K := \text{Res}_{s=-j/2} \Gamma(s) \sum_{k=1}^K \lambda_k^{-s}$ satisfies $|\widehat{b}_j^K - b_j^{\text{Dir}}| \leq C_R K^{-1/2-j/2}$.

Proof. Mellin transform of $Z_\Omega^{\text{Dir}}(t) = \sum_k e^{-\lambda_k t}$ relates spectral coefficients to heat-trace asymptotic coefficients via residues. Truncation error bounded by Weyl-law tail control [13, §4]. $\square$

Cost: $O(KJ)$ for $J = O(\log K)$ coefficients.

9.3. Stage B: Heat coefficients to invariant ring. The Berry coefficients $\{b_j^{\text{Dir}}\}$ are universal polynomial expressions in $\mathcal{I}(\Gamma)$ (Theorem 5.5). The leading expansion is

$$\begin{aligned} b_0^{\text{Dir}} &= \frac{1}{12\pi} I_1^{\text{smooth}} + \sum_l \frac{\pi^2 - \alpha_l^2}{24\pi\alpha_l}, \\ b_1^{\text{Dir}} &= \frac{1}{32\sqrt{\pi}} I_2 + (\text{corner trig.}), \\ b_2^{\text{Dir}} &= a_2^{(0,2)} I_{0,2} + a_2^{(2,1)} I_{2,1} + a_2^{(4)} I_4 + (\text{corner trig.}). \end{aligned}$$

Theorem 9.2. *The map $\mathcal{I}_{\text{star}}^{\leq J}(\Gamma) \rightarrow \mathbb{R}^{J+1}$, $I \mapsto (b_0^{\text{Dir}}, \dots, b_J^{\text{Dir}})$, is a triangular linear isomorphism for every J , invertible by forward substitution.*

Cost: $O(J^2)$ for forward substitution.

9.4. Stage C: Invariants to curvature and angles. Corner angles. The corner-localized invariants $T_j = \sum_l p_j(\alpha_l)$ are symmetric power sums in the corner angles. Newton's identities recover the multiset $\{\alpha_l\}$ via elementary symmetric polynomial inversion.

Smooth-arc curvature. On each real-analytic smooth arc, the curvature Fourier coefficients $\hat{\kappa}_n$ are determined by the integrated curvature invariants via the Edward / Polterovich–Sher triangular recurrence. The leading Polterovich–Sher formula is $a_1(\gamma) = \frac{2}{3} \sum_n (n^3 - n) |\hat{\kappa}_n|^2$ with higher invariants supplying phase data.

Theorem 9.3. *For Γ with real-analytic smooth arcs and corner angles in $(0, 2\pi) \setminus \{\pi\}$, Stage C recovers $\{\kappa_l\}$ on each arc (as a real-analytic function) and $\{\alpha_l\}$ (as a multiset) from $\mathcal{I}(\Gamma)$.*

9.5. Stage D: Frenet integration with corner jumps.

Theorem 9.4. *Given $(\kappa_l, \alpha_l, L_l)$ from Stage C, piecewise integration of the Frenet equations $\theta'(s) = \kappa(s)$ on each arc, with jump $\theta(s_l^+) = \theta(s_l^-) + (\pi - \alpha_l)$ at each corner, produces the boundary curve Γ up to rigid motion. The Gauss–Bonnet identity $\sum_l (\pi - \alpha_l) + \int^{\text{smooth}} \kappa ds = 2\pi$ forces closure.*

9.6. Stage E: Verification via corner moments. The reconstructed corner positions $\{\hat{v}_l\}$ give directed inverse-square moments $\sigma_2^{(l)} = \sum_{l' \neq l} (\hat{v}_l - \hat{v}_{l'})^{-2}$ as a consistency check against the predictions from Stage A–D. Disagreement triggers a small nonlinear least-squares refinement.

9.7. End-to-end error.

Theorem 9.5. *For input spectrum $(\lambda_1, \dots, \lambda_K)$ from $(\Omega, z_0) \in \mathcal{S}_R$ with real-analytic smooth arcs of curvature analytic radius τ , the algorithm produces $\hat{\Omega}$ with*

$$\|\hat{\Gamma} - \Gamma\|_{C^0} \leq C_R K^{-1/2} (\log K)^c,$$

with optimal Fourier truncation $M^ \asymp \tau^{-1} \log K$ balancing analytic tail and data-induced error.*

10. PROOF OF THEOREM B: SHARPNESS VIA Q-CONSTRUCTION

We construct an isospectral non-isometric partner for every non-star-shaped piecewise smooth simply connected planar domain with sufficient piece-multiplicity.

10.1. Forward direction: Sunada-style construction.

Theorem 10.1. *Let Ω be a bounded piecewise smooth simply connected planar domain with boundary decomposition $\partial\Omega = \gamma_1 \cup \dots \cup \gamma_m$. Suppose there exists a permutation $\pi \in \text{Sym}(m)$ such that:*

- (a) *The cyclic reassembly Γ_π forms a closed Jordan curve enclosing a domain Ω_π .*
- (b) *$\mathcal{I}(\Gamma_\pi) = \mathcal{I}(\Gamma)$.*
- (c) *$|\Omega_\pi| = |\Omega|$.*

Then $\text{Spec}(-\Delta_{\Omega_\pi}) = \text{Spec}(-\Delta_\Omega)$.

Proof. The Berry expansion of $Z_\Omega^{\text{Dir}}(t)$ has coefficients in $\mathcal{I}(\Gamma)$ (Theorem 5.5), plus the leading area and perimeter terms. Hypotheses (a)-(c) ensure that all heat-trace coefficients of Ω_π match those of Ω to all orders, hence $Z_{\Omega_\pi}^{\text{Dir}}(t) = Z_\Omega^{\text{Dir}}(t)$ as asymptotic series, hence the spectra agree (Tauberian inversion of Dirichlet series). $\square$

10.2. Reverse direction: star-shaped forces trivial π .

Theorem 10.2. *Let $(\Omega, z_0) \in \mathcal{S}_R$ be star-shaped. For any piecewise smooth simply connected Ω' with $\text{Spec}(-\Delta_{\Omega'}) = \text{Spec}(-\Delta_\Omega)$ and Ω' obtained from Ω by a piece-permutation π as in Theorem 10.1, Ω' is similar to Ω .*

Proof. The polar parametrization from z_0 orders the boundary pieces canonically by angular position θ_l . Cross-arc correlation invariants in $\mathcal{I}(\Gamma)$ — terms involving $\int \int \kappa(s)\kappa(s')|s-s'|^{-2} ds ds'$ from the Q heat-trace expansion — depend on the cyclic ordering of pieces, not just on the multi-set. Preservation of $\mathcal{I}(\Gamma)$ forces π to preserve cyclic order up to rotation; orientation-preservation forces π to be a rotation. Hence $\Omega' = g \cdot \Omega$ for $g \in \text{Sim}(\mathbb{R}^2)$. $\square$

10.3. Gordon-Webb-Wolpert worked example. The 7-triangle propellers [8, 1] admit boundary symbol group $G = \text{PSL}(3, 2) = L_3(2)$ (order 168), with Sunada almost-conjugate non-conjugate subgroups $H_1, H_2 \cong \text{Sym}(4)$ as point-and-line stabilizers in the Fano plane. The induced piece-permutation π on the 9-edge boundary set is a product of three transpositions exchanging the orbit structure. By Theorem 10.1, the two propellers are isospectral; by direct computation (Section 11), they are not star-shaped from any interior point, consistent with Theorem 10.2.

11. TOPOLOGICAL CHARACTERIZATION: THE WINDING-NUMBER ARGUMENT

11.1. Radial projection and degree.

Lemma 11.1. *For Ω bounded with C^1 boundary and $z_0 \in \Omega$, the radial projection*

$$\pi_{z_0}: \partial\Omega \rightarrow S^1, \quad \pi_{z_0}(w) = \frac{w - z_0}{|w - z_0|}$$

has degree 1.

Proof. The degree equals the winding number of $\partial\Omega$ around $z_0 \in \Omega$, which is 1 for a positively-oriented Jordan curve enclosing z_0 . $\square$

Proposition 11.2. *Ω is star-shaped from z_0 if and only if π_{z_0} has no critical points.*

11.2. The FTA-style count.

Proposition 11.3. *For non-star-shaped Ω with C^1 boundary, π_{z_0} has at least 4 critical points.*

Proof. Step 1. Each critical point is non-degenerate (generic case) with signed index ± 1 , depending on whether the radial projection switches from forward-to-backward or backward-to-forward as the boundary is traversed.

Step 2. The degree of π_{z_0} equals the signed alternating sum of critical-point indices, normalized to give degree 1. With k_+ positive and k_- negative critical points,

$$(k_+ - k_-)/2 = \deg(\pi_{z_0}) = 1,$$

so $k_+ = k_- + 2$.

Step 3. Non-star-shaped requires at least one fold, hence $k_- \geq 1$. Total $k_+ + k_- = 2k_- + 2 \geq 4$. $\square$

Remark 11.4. This is the topological analog of the FTA-via-winding-number argument. A polynomial of degree $n \geq 1$ has at least n zeros because the winding number of its image around a large circle equals n ; here, the radial projection's degree 1 combined with non-trivial fold structure forces ≥ 4 critical points. The argument is the same structural principle: degree is a topological invariant computable as a signed preimage count, and constraints on the degree force lower bounds.

11.3. Implication for piece-multiplicity.

Corollary 11.5. *Every non-star-shaped bounded piecewise smooth simply connected planar domain admits, for every interior point z_0 , a decomposition of $\partial\Omega$ into at least 4 pieces, each locally star-shaped from z_0 .*

The piece-multiplicity condition in Theorem 10.1 is therefore generic in the non-star-shaped class. With ≥ 4 pieces, sufficient combinatorial freedom exists for the construction of a non-trivial $\mathcal{I}$ -preserving permutation π . For polygonal domains with ≥ 4 reflex corners and sufficient combinatorial freedom in the gluing, the Sunada construction supplies π explicitly; for general piecewise smooth domains, perturbative deformation to polygonal approximation gives π as a limit.

Remark 11.6. The GWW propellers have three reflex corners arranged symmetrically; by Proposition 11.3, the radial projection from any interior point has at least 4 critical points (in fact, 6 — two for each reflex corner). The propellers are not star-shaped from any interior point.

12. PROOF OF THEOREM C: ISOMETRY RIGIDITY VIA HADAMARD DATA

The third theorem applies to the full piecewise smooth class — without the star-shaped restriction — at the cost of requiring the Hadamard second variation $D^2\lambda_1$ as auxiliary data.

12.1. Hadamard formula.

Proposition 12.1. *For Ω piecewise smooth simply connected with simple first Dirichlet eigenvalue λ_1 and eigenfunction u_1 ($\|u_1\|_{L^2} = 1$), and for boundary normal velocity $V \in C^{1,\alpha}(\partial\Omega)/\mathbb{R}$,*

$$D^2\lambda_1|_{\Omega}[V] = -2 \sum_{k \geq 2} \frac{|\langle V \partial_\nu u_1, \partial_\nu u_k \rangle_{\partial\Omega}|^2}{\lambda_k - \lambda_1} + \int_{\partial\Omega} \kappa(s) V(s)^2 (\partial_\nu u_1(s))^2 ds.$$

12.2. Kernel decomposition. Polarizing $D^2\lambda_1$ to a bilinear form $\mathcal{H}(V, W)$ on $L^2(\partial\Omega)$ and reading as an integral kernel on $\partial\Omega \times \partial\Omega$:

$$H(x, y) = -2(\partial_\nu u_1)(x)(\partial_\nu u_1)(y) G_R(x, y) + \kappa(x)\delta(x - y)(\partial_\nu u_1(x))^2,$$

with

$$G_R(x, y) := \sum_{k \geq 2} \frac{(\partial_\nu u_k)(x)(\partial_\nu u_k)(y)}{\lambda_k - \lambda_1} = \partial_\nu^x \partial_\nu^y (-\Delta_\Omega - \lambda_1)^{-1}_{\perp u_1}(x, y) \Big|_{\partial\Omega \times \partial\Omega}.$$

12.3. Boundary symbol: the dressed chord Laplacian.

Proposition 12.2. *On the smooth part of $\partial\Omega$, G_R is a classical pseudodifferential operator of order $+1$. Its principal symbol is $|\xi|$, equivalently its kernel has the Hadamard finite-part expansion*

$$G_R(s, s + \epsilon) = -\frac{1}{\pi} \text{f.p.} \frac{1}{|\gamma(s + \epsilon) - \gamma(s)|^2} + R_0(s, s + \epsilon)$$

in arc-length coordinates, where R_0 is the kernel of an operator of order at most 0. The shift by λ_1 and the removal of the ground-state pole affect only lower-order, smoothing, or finite-rank terms and leave the principal symbol unchanged.

Proof. In the half-plane model, the Dirichlet resolvent has order -2 in the interior; applying one normal derivative at each boundary variable raises the boundary operator by two orders, producing an order $+1$ boundary pseudodifferential operator. Its principal symbol is the same $|\xi|$ as the Dirichlet-to-Neumann operator. Curving the boundary changes only lower-order terms by the standard boundary calculus, while replacing $-\Delta$ by $-\Delta - \lambda_1$ does not change the high-frequency symbol. Excluding u_1 is a finite-rank regularization of the resolvent pole and therefore does not alter the principal part. $\square$

Thus

$$H_{\text{res}}(s, t) = -2p(s)p(t)G_R(s, t), \quad p = \partial_\nu u_1,$$

is a *dressed chord Laplacian*: up to lower-order terms,

$$H_{\text{res}} = -\frac{2}{\pi} M_p Q_\Gamma M_p + \Psi^0,$$

where M_p denotes multiplication by p and the sign is fixed by the convention that Q_Γ has off-diagonal finite-part kernel $-|x - y|^{-2}$.

12.4. Spectral subtraction.

Lemma 12.3. *The decomposition $H = H_{\text{curv}} + H_{\text{res}}$ separates canonically as Dirac-supported H_{curv} and finite-part pseudodifferential H_{res} . From $\mathcal{H}(V, W)$ as V, W range over $C^\infty(\partial\Omega)$:*

- (i) $H_{\text{curv}}(x, x) = \kappa(x)(\partial_\nu u_1(x))^2$ is recovered as a function on the smooth boundary plus Dirac masses at corners.
- (ii) H_{res} is recovered as an order-one finite-part distribution on $\partial\Omega \times \partial\Omega$.

12.5. Pointwise recovery of $\partial_\nu u_1$.

Proposition 12.4. *From the kernel H and the spectrum $\{\lambda_k\}$, the function $\partial_\nu u_1(x)$ is recovered pointwise on the smooth interior of $\partial\Omega$, up to global sign (fixed positive by the Hopf maximum principle).*

Proof. The Hopf principle gives $p := \partial_\nu u_1 > 0$ on the smooth boundary. By Proposition 12.2,

$$H_{\text{res}}(s, s + \epsilon) = \frac{2}{\pi} p(s) p(s + \epsilon) \text{ f. p. } \frac{1}{|\gamma(s + \epsilon) - \gamma(s)|^2} + O(\epsilon^{-1})$$

in the finite-part Laurent expansion. Since

$$\frac{1}{|\gamma(s + \epsilon) - \gamma(s)|^2} = \epsilon^{-2} + \frac{\kappa(s)^2}{12} + O(\epsilon),$$

the leading pole isolates the square of the ground-state flux:

$$p(s)^2 = \frac{\pi}{2} \text{Coef}_{\epsilon^{-2}} [H_{\text{res}}(s, s + \epsilon)].$$

The global sign is fixed by Hopf positivity. The Rellich–Pohozaev identity

$$\int_{\partial\Omega} p(s)^2 (x \cdot \nu) ds = 2\lambda_1$$

gives an independent normalization check. $\square$

12.6. Higher fluxes and boundary spectral data.

Theorem 12.5. *From $\{\lambda_k\}$ and $D^2\lambda_1$, the full sequence of boundary fluxes $\{\partial_\nu u_k\}_{k \geq 1}$ is recovered pointwise on the smooth boundary (up to global sign per simple eigenmode, up to orthogonal mixing in multiplicity spaces).*

Proof. Proposition 12.4 recovers $\partial_\nu u_1$. Dividing the finite-part kernel by the recovered multiplication weights gives

$$G_R(x, y) = -\frac{H_{\text{res}}(x, y)}{2(\partial_\nu u_1)(x)(\partial_\nu u_1)(y)}$$

as a known order-one boundary pseudodifferential operator. Its spectral resolution is

$$G_R = \sum_{k \geq 2} \frac{|\partial_\nu u_k\rangle \langle \partial_\nu u_k|}{\lambda_k - \lambda_1},$$

and the eigenvalue gaps $\lambda_k - \lambda_1$ are known from the spectrum. Diagonalising G_R recovers each $\partial_\nu u_k$ pointwise on the smooth boundary, up to sign for simple eigenmodes and up to orthogonal mixing in multiplicity spaces. $\square$

12.7. Boundary control method.

Proposition 12.6. *The boundary spectral data $\{\lambda_k, \partial_\nu u_k|_{\partial\Omega}\}$ determines the Dirichlet-to-Neumann map $\Lambda_{\partial\Omega}$ on $\partial\Omega$, and conversely.*

Theorem 12.7 (Belishev-Kurylev). *For piecewise smooth Ω_1, Ω_2 , equivalence of DtN maps under a boundary diffeomorphism implies isometry of the domains [2, 20, 30].*

12.8. Proof of Theorem 1.3.

Proof of Theorem 1.3. By Theorem 12.5, the boundary spectral data for Ω_i is determined by $(\text{Spec}(-\Delta_{\Omega_i}), D^2\lambda_1|_{\Omega_i})$. The Hadamard form isomorphism (hypothesis (ii)) induces a unitary intertwiner between the boundary L^2 spaces preserving the spectral data, hence (Proposition 12.6) the DtN maps. By Theorem 12.7, $\Omega_2 = g \cdot \Omega_1$ for $g \in \text{Iso}(\mathbb{R}^2)$. $\square$

12.9. Breaking the GWW degeneracy. The Sunada transplantation map $T: L^2(\Omega_1) \rightarrow L^2(\Omega_2)$ for the GWW propellers intertwines Dirichlet Laplacians but is a combinatorial reshuffling of triangular tiles, not a pointwise boundary diffeomorphism. The boundary fluxes $T(\partial_\nu u_1^{(1)})$ and $\partial_\nu u_1^{(2)}$ are different functions on $\partial\Omega_2$, and the Hadamard kernels $H^{(1)}, H^{(2)}$ are not isomorphic as bilinear forms.

Proposition 12.8. $D^2\lambda_1|_{\Omega_1}$ and $D^2\lambda_1|_{\Omega_2}$ are not isomorphic as quadratic forms for the BCDS propellers Ω_1, Ω_2 . The pair $(\text{Spec}, D^2\lambda_1)$ correctly distinguishes them.

13. THE COMPLETE CHARACTERIZATION

The three theorems together form a complete characterization of geometric audibility on the bounded piecewise smooth simply connected class:

Spectrum alone is sufficient on $\mathcal{S}_R$ (Theorem 1.1). Star-shapedness is the structural feature that lets the spectrum alone determine the domain up to similarity. The proof factors through the Q -operator and its heat-trace bridge to the curvature-angle invariant ring $\mathcal{I}(\Gamma)$, with cyclotomic rigidity supplying the local injectivity at the regular attractor of each cyclic-symmetric stratum.

Spectrum alone fails outside $\mathcal{S}_R$ (Theorem 1.2). For generic non-star-shaped domains, there exists a Q -construction partner — a piece-permutation of the boundary preserving $\mathcal{I}(\Gamma)$ together with area and perimeter, yielding an isospectral non-isometric reassembly. The GWW propellers are the canonical worked example. The piece-multiplicity required for this construction is generic in the non-star-shaped class by the winding-number topological argument (Theorem 1.4).

Spectrum plus Hessian determines isometry universally (Theorem 1.3). The Hadamard second variation $D^2\lambda_1$ encodes the full boundary spectral data $\{\partial_\nu u_k|_{\partial\Omega}\}$ via spectral subtraction. The key microlocal fact is that the boundary-restricted double-normal reduced resolvent is not logarithmic; it is an order-one finite-part operator with the same leading hypersingular kernel as Q_Γ . The Hessian is therefore a multiplication-dressed chord Laplacian, and the ϵ^{-2} Laurent coefficient recovers $(\partial_\nu u_1)^2$ algebraically. Combined with the spectrum itself, this is equivalent to the Dirichlet-to-Neumann map, hence (Belishev-Kurylev) determines the domain up to isometry on the full piecewise smooth class.

Class	Data	Determination
Star-shaped $\mathcal{S}_R$	$\text{Spec}(-\Delta_\Omega)$ alone	Up to similarity (Thm. 1.1)
Non-star-shaped (generic)	$\text{Spec}(-\Delta_\Omega)$ alone	Fails (Thm. 1.2)
Full piecewise smooth	$\text{Spec}(-\Delta_\Omega), D^2\lambda_1$	Up to isometry (Thm. 1.3)

The trichotomy is structurally sharp: the spectrum alone is sufficient on a class that is both natural (star-shaped is the maximal class on which the polar parametrization makes sense) and characterized topologically (no critical points of radial projection); the Hadamard Hessian is the minimal additional spectral data needed to break the GWW obstruction outside this class.

13.1. Connections to deeper mathematics. The trichotomy connects to several distinct mathematical structures:

Cyclotomic rigidity and the Stark regulator. The rate of convergence in the contraction flow is $\log(2\eta_{n'}) = \log 2 + L'(0, \chi_{n'})$, with the Stark regulator $\log \eta_{n'}$ supplying the arithmetic content. For the regular pentagon ($n' = 5$): $\log \varphi = L'(0, \chi_5)$ is the Stark regulator of $\mathbb{Q}(\sqrt{5})$ [4, 28].

Boundary control method. Theorem 1.3 reduces the inverse spectral problem on bounded domains to the inverse boundary problem (Belishev-Kurylev, Calderón-Sylvester-Uhlmann). The Hadamard Hessian acts as a microlocal bridge extracting boundary spectral data from interior spectral data: its reduced resolvent component has the same principal symbol as the Steklov/chord operator, while the ground-state flux appears as the dressing multiplier.

Topological foundation. The winding-number argument (Theorem 1.4) is the inverse-spectral analog of the FTA via winding numbers: both use the degree of a circle map as a topological invariant forcing lower bounds on preimage counts.

Operator-canonical fractional iterates. The contraction flow uses the golden tetratation base $b_\star = \exp(1/(2\varphi e^{1/(2\varphi)}))$, the unique base for which the operator-canonical fractional iterate has multiplier $1/(2\varphi)$. This construction [14] ties the inverse-spectral problem to complex tetratation and Schröder linearization.

14. OPEN PROBLEMS AND OBSTRUCTIONS

14.1. Non-cyclic strata. The unconditional version of Theorem 1.1 on cyclic-symmetric strata requires extension to non-cyclic strata. The contraction-flow conjecture (Section 7) supplies the missing piece: every stratum should admit a real-analytic flow converging to its higher-symmetry boundary. The conjecture is open.

14.2. Conjecture 14.2. The reverse direction of the star-shaped characterization — whether every isospectral counterexample in the simply connected planar class must be non-star-shaped — is open. Theorem 1.2 establishes the generic-density version (every non-star-shaped domain in a generic class admits a partner), but the unconditional converse (every isospectral counterexample is non-star-shaped) requires showing that no Sunada-style construction produces a star-shaped pair. We conjecture this is true.

Conjecture. Every pair of bounded simply connected planar domains with identical Dirichlet spectra and distinct shapes contains at least one member that is not star-shaped from any interior point.

14.3. Cusps and reflex limits. A cardioid from any non-cusp center has r bounded but the boundary still contains a cusp; the chord-arc inequality fails. The framework requires a renormalized Q operator with cusp-specific asymptotics; the Steklov side is treated in [23].

14.4. Multiply connected domains. An annular domain has two boundary components. The chord-arc identity and heat-trace bridge extend; the cyclic-symmetric attractor is the round annulus, and the rigidity rate involves the modular parameter (ratio of two radii). A natural extension; multi-component Mellin calculus is the new technical work.

14.5. Higher dimensions. In dimension $n \geq 3$, the BC-method and Calderón inverse problem are more delicate. The framework of Theorems 1.1–1.3 likely extends with care; the Calderón problem in higher dimensions is itself a major open area [19].

14.6. Numerical implementation of the Hadamard recovery. The pointwise recovery of $\partial_\nu u_1$ from the Hadamard form (Proposition 12.4) is algebraic at the continuum level: extract the ϵ^{-2} coefficient of the finite-part Laurent expansion of $H_{\text{res}}(s, s + \epsilon)$. A stable numerical implementation still requires robust finite-part coefficient extraction near corners and near Kondrat'ev singularities, especially for reflex angles where boundary fluxes may have low regularity. This remains open.

APPENDIX A. FINITE-DIMENSIONAL ALGEBRAIC CORE

This appendix records the finite-dimensional mechanism underlying the polygonal and low-mode parts of the paper. It is the common algebraic core of the convex-polygon and resolved star-shaped arguments.

A.1. Directed inverse-square moments. Let P be a polygon with distinct vertices $z_0, \dots, z_{n-1} \in \mathbb{C}$. Set

$$\sigma_2^{(i)}(P) := \sum_{j \neq i} (z_i - z_j)^{-2}$$

and impose the product-normalised similarity gauge

$$\sum_k z_k = 0, \quad \prod_k z_k = (-1)^n, \quad \arg z_0 = 0.$$

The resulting moduli cell $\mathcal{M}_n$ has real dimension $2n - 4$.

Theorem A.1 (Algebraic inverse theorem). *For every $n \geq 3$, the map*

$$\Phi: \mathcal{M}_n \rightarrow \mathbb{C}^n, \quad P \mapsto (\sigma_2^{(0)}(P), \dots, \sigma_2^{(n-1)}(P))$$

is globally injective. The inverse is algebraic, obtained from the Cauchy field

$$\psi_i(z) = \sum_{j \neq i} (z - z_j)^{-1}$$

by Newton–Girard recovery of the pole positions.

Proof. The complex Jacobian $M_{ik} = \partial \sigma_2^{(i)} / \partial z_k$ has rank at least $n - 1$. Indeed, if $v \in \ker M$, define $f(z) = \prod_k (z - z_k)$ and $F(z) = f(z)(\sum_k v_k (z - z_k)^{-1})^2$. Then $\deg F \leq 2n - 2$, while the kernel condition imposes $2n - 1$ value-and-derivative constraints. The solution space is therefore at most one-dimensional and contains the translation vector $\mathbf{1}$. The centroid, product, and rotation gauges are transverse to this translation kernel, so $D_{\mathbb{R}}\Phi$ has full rank on $T_P\mathcal{M}_n$.

Properness follows because $|\sigma_2^{(i)}| \rightarrow \infty$ when two vertices collide. The moduli cell is contractible. Hadamard–Cacciopoli’s global inverse theorem [26] therefore gives global injectivity. The rational inverse follows by expanding ψ_i at z_i ; the Taylor coefficients are symmetric functions of $(z_i - z_j)^{-1}$, and Newton–Girard recovers the unordered pole set. $\square$

A.2. The inverse-square operator. For vertices z_i , define

$$(Q_P)_{ij} = \begin{cases} -|z_i - z_j|^{-2}, & i \neq j, \\ \sum_{k \neq i} |z_i - z_k|^{-2}, & i = j. \end{cases}$$

Then Q_P is the Hermitian Hessian of the logarithmic discriminant:

$$Q_{ij} = \frac{\partial^2}{\partial z_i \partial \bar{z}_j} \sum_{a < b} \log |z_a - z_b|^2.$$

Consequently $Q_P \succeq 0$ on $\mathbf{1}^\perp$ and has row sums zero. In cyclic coordinates, with $\phi_k^{(m)} = e^{2\pi i m k / n}$ and

$$G(d) = \sum_{k=0}^{n-1} |z_k - z_{k+d}|^{-2},$$

one has the mode–geometry identity

$$R_m := \frac{\langle \phi^{(m)}, Q_P \phi^{(m)} \rangle}{\|\phi^{(m)}\|^2} = \frac{1}{n} \sum_{d=1}^{n-1} (1 - \cos(2\pi m d / n)) G(d).$$

The cosine blocks are invertible after decomposing by divisors of n , so the cyclic mode energies determine the distance shell data and hence Q_P .

A.3. Bessel bridge and quantitative chart. At the disk, $u_1(r) = cJ_0(j_{0,1}r)$ and the second variation under $V(\theta) = \cos(m\theta)$ satisfies

$$\left. \frac{d^2(\lambda_1 \text{Area})}{d\varepsilon^2} \right|_{D,m} = 2\pi c_0 \left[(m+1) - \frac{j_{0,1}J_{m+1}(j_{0,1})}{J_m(j_{0,1})} \right],$$

where $c_0 = (\partial_\nu u_1)^2|_{\partial D} = j_{0,1}^2/\pi$. The $m=1$ term vanishes because it is the translation gauge. For $m \geq 2$, the associated isotype multipliers

$$\kappa_m^{\text{disk}} = \frac{4\pi^2 c_0}{m} \left[(m+1) - \frac{j_{0,1}J_{m+1}(j_{0,1})}{J_m(j_{0,1})} \right]$$

are uniformly bounded away from zero; numerically they lie in the interval $[68.7, 80.0]$ and tend to $4\pi j_{0,1}^2$.

For a C^2 finite-dimensional family $\theta \mapsto \Omega_\theta$, let

$$F_K(\theta, \ell) = (\lambda_1(e^\ell \Omega_\theta), \dots, \lambda_K(e^\ell \Omega_\theta)).$$

If $\sigma_F = \inf \sigma_{\min}(DF_K) > 0$ and $M_F = \sup \|D^2 F_K\|_{\text{op}}$ on a ball with radius $< \sigma_F/M_F$, then any residual $E < \sigma_F^2/(2M_F)$ gives

$$\|\hat{p} - p_*\| \leq \frac{\sigma_F - \sqrt{\sigma_F^2 - 2M_F E}}{M_F} \leq \frac{2E}{\sigma_F}.$$

This is the quantitative estimate used in the polygonal, resolved smooth, and mixed-stratum reconstructions.

APPENDIX B. RESOLVED RECONSTRUCTION ALGORITHM

This appendix expands Section 9. The algorithm is the constructive form of the hearing chain

$$\text{Spec}(-\Delta_\Omega) \longrightarrow \{b_j^{\text{Dir}}\} \longrightarrow \mathcal{I}(\Gamma) \longrightarrow (\kappa, \{\alpha_l\}) \longrightarrow \Gamma.$$

B.1. Stage A: spectral zeta residues. The spectral zeta function

$$\zeta_\Omega^{\text{Dir}}(s) = \sum_{k \geq 1} \lambda_k^{-s}$$

has Mellin relation

$$\Gamma(s) \zeta_\Omega^{\text{Dir}}(s) = \int_0^\infty t^{s-1} Z_\Omega^{\text{Dir}}(t) dt.$$

Thus

$$|\Omega| = 4\pi \text{Res}_{s=1} \zeta_\Omega^{\text{Dir}}(s), \quad |\Gamma| = -8\sqrt{\pi} \text{Res}_{s=1/2} \zeta_\Omega^{\text{Dir}}(s),$$

and, for $j \geq 0$,

$$b_j^{\text{Dir}}(\Gamma) = \text{Res}_{s=-j/2} \Gamma(s) \zeta_\Omega^{\text{Dir}}(s).$$

For truncated input $(\lambda_1, \dots, \lambda_K)$, Weyl-tail control gives

$$|\hat{b}_j^K - b_j^{\text{Dir}}| \leq C_R K^{-1/2-j/2}.$$

B.2. Stage B: triangular inversion in the invariant ring. The curvature-angle ring is

$$\mathcal{I}(\Gamma) = \mathbb{R} \left[\sum_l \int_{\gamma_l} \kappa^{i_0} (\kappa')^{i_1} \dots (\kappa^{(k)})^{i_k} ds \right] / \text{IBP} \oplus \mathbb{R} \left[\sum_l p(\alpha_l) \right],$$

where p ranges over trigonometric rational polynomials generated by $\sin, \cos, \cot, \csc$. The first Berry coefficients have the form

$$\begin{aligned} b_0^{\text{Dir}} &= \frac{1}{12\pi} \int^{\text{smooth}} \kappa ds + \sum_l \frac{\pi^2 - \alpha_l^2}{24\pi\alpha_l}, \\ b_1^{\text{Dir}} &= \frac{1}{32\sqrt{\pi}} I_2 + (\text{corner trigonometric terms}), \\ b_2^{\text{Dir}} &= a_2^{(0,2)} I_{0,2} + a_2^{(2,1)} I_{2,1} + a_2^{(4)} I_4 + (\text{corner trigonometric terms}). \end{aligned}$$

The system is triangular by weight: b_j^{Dir} contains only invariants of weight at most j , and its top-weight coefficient is non-zero. Therefore the finite truncation $\mathcal{I}^{\leq J} \rightarrow (b_0, \dots, b_J)$ is inverted by forward substitution.

B.3. Stage C: curvature and angles. Corner angles are recovered from symmetric power sums

$$T_q = \sum_l q(\alpha_l),$$

using Newton's identities. A monotone one-variable generator such as

$$q(\alpha) = \frac{\pi^2 - \alpha^2}{24\pi\alpha}$$

recovers the multiset $\{\alpha_l\}$ after root-finding.

On each real-analytic arc, Edward–Polterovich–Sher inversion recovers curvature Fourier coefficients from integrated invariants. The leading Steklov coefficient has the model form

$$a_1(\gamma) = \frac{2}{3} \sum_{n \geq 2} (n^3 - n) |\widehat{\kappa}_n|^2,$$

and higher derivative-containing invariants recover phases. Real analyticity gives $|\widehat{\kappa}_n| \leq C_\Gamma e^{-\tau n}$.

B.4. Stage D: Frenet integration. Given κ_l on arcs and corner angles α_l , integrate

$$\theta'(s) = \kappa(s), \quad \gamma'(s) = e^{i\theta(s)}$$

on each arc, with corner jump

$$\theta(s_l^+) = \theta(s_l^-) + (\pi - \alpha_l).$$

Gauss–Bonnet,

$$\sum_l (\pi - \alpha_l) + \int_\Gamma^{\text{smooth}} \kappa ds = 2\pi,$$

forces the total turning. Closure $\gamma(L) = \gamma(0)$ fixes the remaining similarity gauge. If $\|\widehat{\kappa} - \kappa\|_{C^0} \leq \varepsilon$, then

$$\|\widehat{\Gamma} - \Gamma\|_{C^0} \leq C_R L^2 \varepsilon.$$

B.5. Stage E: corner-moment verification. For reconstructed corners $\widehat{v}_l$, compute

$$\widehat{\sigma}_2^{(l)} = \sum_{l' \neq l} (\widehat{v}_l - \widehat{v}_{l'})^{-2}.$$

The same directed moments are predicted by the polygonal skeleton of the Q heat-trace data. Agreement is a consistency check; disagreement triggers a local least-squares refinement of corner positions against the predicted $\sigma_2^{(l)}$ values.

B.6. Error balance. For analytic radius τ , truncating curvature at mode M gives

$$\|\widehat{\kappa}_{\leq M} - \kappa\|_{C^0} \leq C_\Gamma \left(e^{-\tau M} + K^{-1/2} M^c \right).$$

Choosing $M \asymp \tau^{-1} \log K$ yields the global estimate

$$d_H(\partial \widehat{\Omega}_K, \partial \Omega) \leq C_R K^{-1/2} (\log K)^c$$

modulo similarity.

APPENDIX C. SHARPNESS AND SUNADA/Q REASSEMBLY

C.1. Heat-trace agreement. Let Γ_1, Γ_2 be piecewise smooth Jordan curves bounding Ω_1, Ω_2 . If

$$|\Omega_1| = |\Omega_2|, \quad |\Gamma_1| = |\Gamma_2|, \quad \mathcal{I}(\Gamma_1) = \mathcal{I}(\Gamma_2),$$

then all Berry coefficients agree. Hence the Dirichlet heat traces have the same full asymptotic expansion. In the real-analytic class this recovers the heat trace by Borel summability; in the general piecewise smooth setting the equality is interpreted through the meromorphic spectral zeta function and a Tauberian uniqueness theorem for Dirichlet series [16].

C.2. Boundary-piece permutations. Let $\Gamma = \gamma_1 \cup \cdots \cup \gamma_m$ with corner angles α_l . A permutation $\pi \in \text{Sym}(m)$ gives a reassembly

$$\Gamma_\pi = \gamma_{\pi(1)} \cup \cdots \cup \gamma_{\pi(m)}$$

provided the endpoints and tangent jumps close to a Jordan curve. It preserves $\mathcal{I}(\Gamma)$ if it preserves:

- (1) the multiset of arc curvature functions;
- (2) the multiset of corner angles;
- (3) all cross-arc correlation terms appearing in the Q heat trace.

For star-shaped domains the polar order is canonical, so these cross-correlation terms force π to be a cyclic rotation or reflection; orientation and signed-area positivity remove the reflection unless it is an isometry.

C.3. Sunada symbol groups. A boundary symbol group is a finite group G acting on the boundary piece set and preserving local piece data. If $H_1, H_2 \leq G$ are Sunada almost-conjugate,

$$|H_1 \cap [g]| = |H_2 \cap [g]| \quad (g \in G),$$

then character sums of all class functions agree. Every G -equivariant expression in $\mathcal{I}(\Gamma)$ is such a character sum; hence the two orbit reassemblies have identical area, perimeter, and $\mathcal{I}$ data, and are isospectral.

For the Gordon–Webb–Wolpert/BCDS propellers, $G = \text{PSL}(3, 2)$ acts through the Fano-plane labelling of the seven triangular tiles. The two subgroups are point and line stabilizers, both isomorphic to $\text{Sym}(4)$, almost-conjugate but not conjugate. The induced boundary permutation exchanges the relevant edge orbits while preserving the angle and edge-length multisets. The three reflex corners force failure of star-shapedness from every interior point.

C.4. Genericity scope. The radial projection $\pi_{z_0}: \partial\Omega \rightarrow S^1$ has degree 1. If Ω is not star-shaped from z_0 , this circle map has folds. Non-degenerate folds carry signs, and the signed fold count satisfies

$$\frac{k_+ - k_-}{2} = 1.$$

Since a non-star-shaped map has at least one negative fold, $k_+ + k_- \geq 4$. Thus every non-star-shaped domain admits at least a four-piece locally star-shaped decomposition. The exact Sunada/Q partner statement is generic rather than universal: the set of non-star-shaped domains with an exact $\mathcal{I}$ -preserving reassembly is open dense in the piecewise smooth non-star-shaped class, while arbitrary smooth asymmetric examples may only be approximated by such domains.

APPENDIX D. HADAMARD BOUNDARY-CONTROL RECONSTRUCTION

D.1. Moduli tangent spaces and the Hessian. Let $\mathcal{M}$ be the area-normalized moduli space of bounded piecewise $C^{2,\alpha}$ simply connected domains with corner angles in $(0, 2\pi) \setminus \{\pi\}$. The tangent space is identified with boundary normal velocities $V \in C^{1,\alpha}(\partial\Omega)/\mathbb{R}$. For the first normalized eigenfunction u_1 ,

$$D^2\lambda_1[V] = -2 \sum_{k \geq 2} \frac{|\langle V \partial_\nu u_1, \partial_\nu u_k \rangle_{\partial\Omega}|^2}{\lambda_k - \lambda_1} + \int_{\partial\Omega} \kappa V^2 (\partial_\nu u_1)^2 ds,$$

with curvature interpreted as a measure at corners in the Kondrat'ev-Grisvard setting [18, 9].

D.2. Kernel decomposition. Polarization gives a bilinear form $\mathcal{H}(V, W)$ with distribution kernel

$$H(x, y) = -2p(x)p(y)G_R(x, y) + \kappa(x)\delta(x - y)p(x)^2, \quad p = \partial_\nu u_1,$$

where

$$G_R(x, y) = \sum_{k \geq 2} \frac{\partial_\nu u_k(x) \partial_\nu u_k(y)}{\lambda_k - \lambda_1}.$$

The curvature term is a Dirac measure on the diagonal. The reduced boundary resolvent term is not the interior logarithmic Green kernel restricted to the boundary; after applying one normal derivative in each variable it is an order-one boundary pseudodifferential operator. Its diagonal singularity is the Hadamard finite part of an inverse-square chord kernel.

D.3. Boundary symbol and Laurent extraction. In local arc-length coordinates on the smooth boundary,

$$G_R(s, s + \epsilon) = -\frac{1}{\pi} \text{f.p.} \frac{1}{|\gamma(s + \epsilon) - \gamma(s)|^2} + R_0(s, s + \epsilon),$$

where R_0 has order at most 0. The principal symbol is $|\xi|$, the same as the Steklov operator. The shift by λ_1 in the reduced resolvent and the projection away from the ground state change only lower-order, smoothing, or finite-rank terms, so the leading finite-part pole is universal.

The chord denominator has the Frenet/Bernoulli expansion

$$\frac{1}{|\gamma(s + \epsilon) - \gamma(s)|^2} = \epsilon^{-2} + \frac{\kappa(s)^2}{12} + O(\epsilon),$$

with higher coefficients universal polynomials in the boundary jet. In the circle model this is the familiar identity

$$\frac{1}{|1 - e^{i\epsilon}|^2} = \frac{1}{4 \sin^2(\epsilon/2)} = \epsilon^{-2} + \frac{1}{12} + \frac{\epsilon^2}{240} + \frac{\epsilon^4}{6048} + \dots$$

Therefore

$$H_{\text{res}}(s, s + \epsilon) = \frac{2}{\pi} p(s)p(s + \epsilon) \text{ f. p. } \frac{1}{|\gamma(s + \epsilon) - \gamma(s)|^2} + \text{lower-order terms},$$

and the leading Laurent coefficient gives

$$p(s)^2 = \frac{\pi}{2} \text{Coef}_{\epsilon^{-2}} [H_{\text{res}}(s, s + \epsilon)].$$

This is the algebraic replacement for logarithmic diagonal peeling.

D.4. Recovering fluxes. The Hopf principle gives $p > 0$ on the smooth boundary. The off-diagonal kernel

$$K_R(x, y) := -\frac{1}{2} H_{\text{res}}(x, y) = p(x)p(y)G_R(x, y)$$

is a multiplication-dressed chord Laplacian, modulo lower-order pseudodifferential corrections. Positivity of p , the ϵ^{-2} Laurent coefficient of H_{res} , and the Rellich identity

$$\int_{\partial\Omega} p(x)^2 (x \cdot \nu) ds = 2\lambda_1$$

recover p pointwise on the smooth boundary and provide an independent normalization check. The global sign is fixed by Hopf positivity.

Dividing the finite-part kernel by $p(x)p(y)$ recovers $G_R(x, y)$ as an order-one boundary pseudodifferential operator. Since the eigenvalue gaps $(\lambda_k - \lambda_1)^{-1}$ are known from the spectrum, spectral decomposition of G_R recovers each boundary flux $\partial_\nu u_k$ up to sign for simple eigenvalues and up to orthogonal mixing in multiplicity spaces.

D.5. From boundary spectral data to the domain. The boundary spectral data determine the Dirichlet-to-Neumann map by

$$\Lambda f = - \sum_{k \geq 1} \frac{\langle f, \partial_\nu u_k \rangle_{\partial\Omega}}{\lambda_k} \partial_\nu u_k.$$

Belishev–Kurylev boundary control, equivalently the two-dimensional Calderon uniqueness theorem in this Euclidean setting [2, 20, 30], determines the domain up to isometry from the DtN map. Thus the pair $(\text{Spec}(-\Delta_\Omega), D^2\lambda_1)$ determines every bounded piecewise smooth simply connected planar domain up to isometry.

D.6. GWW degeneracy. The Sunada transplantation between the BCDS propellers is a combinatorial tile map, not a pointwise boundary diffeomorphism. It intertwines the Dirichlet Laplacians but does not identify the boundary flux functions. Therefore the Hadamard kernels $H^{(1)}$ and $H^{(2)}$ are not isomorphic as boundary quadratic forms; otherwise Theorem 1.3 would force the non-isometric propellers to be isometric. This is exactly the obstruction that the Hessian removes.

REFERENCES

- [1] P. Buser, J. Conway, P. Doyle, and K.-D. Semmler, *Some planar isospectral domains*, Internat. Math. Res. Notices 9 (1994), 391–400.
- [2] M. I. Belishev, *Boundary control in reconstruction of manifolds and metrics (the BC method)*, Inverse Problems 13 (1997), R1–R45.
- [3] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford Univ. Press, 1959.
- [4] S. Chowla and A. Selberg, *On Epstein's zeta function*, J. Reine Angew. Math. 227 (1967), 86–110.
- [5] M. Costabel and E. P. Stephan, *Boundary integral equations for mixed boundary value problems in polygonal domains and Galerkin approximation*, in: *Mathematical Models and Methods in Mechanics 1981*, PWN, Warsaw, 1985, pp. 175–251.

- [6] J. Edward, *An inverse spectral result for the Neumann operator on planar domains*, J. Funct. Anal. 111 (1993), 312–322.
- [7] O. Giraud and K. Thas, *Hearing shapes of drums: mathematical and physical aspects of isospectrality*, Rev. Modern Phys. 82 (2010), 2213–2255.
- [8] C. Gordon, D. L. Webb, and S. Wolpert, *One cannot hear the shape of a drum*, Bull. Amer. Math. Soc. 27 (1992), 134–138.
- [9] P. Grisvard, *Elliptic Problems in Nonsmooth Domains*, SIAM, Philadelphia, 2011.
- [10] R. Gulati, *Hearing the shape of a convex polygon: the cyclotomic-rigidity proof*, manuscript.
- [11] R. Gulati, *Hearing the shape of a smooth star-shaped curve*, manuscript.
- [12] R. Gulati, *Hearing the shape of a piecewise smooth convex planar domain*, manuscript.
- [13] R. Gulati, *A constructive reconstruction algorithm for the star-shaped hearing theorem*, manuscript.
- [14] R. Gulati, *Operator-canonical fractional iterates and complex tetration*, manuscript.
- [15] A. Henrot and M. Pierre, *Shape Variation and Optimization: A Geometrical Analysis*, EMS Tracts Math. 28, European Math. Soc., 2018.
- [16] G. H. Hardy and W. W. Rogosinski, *Notes on Fourier Series and Dirichlet Series*, Cambridge Univ. Press, 1944.
- [17] M. Kac, *Can one hear the shape of a drum?*, Amer. Math. Monthly 73 (1966), no. 4, part II, 1–23.
- [18] V. A. Kondrat’ev, *Boundary value problems for elliptic equations in domains with conical or angular points*, Trudy Moskov. Mat. Obshch. 16 (1967), 209–292.
- [19] C. E. Kenig, J. Sjöstrand, and G. Uhlmann, *The Calderón problem with partial data*, Ann. of Math. (2) 165 (2007), 567–591.
- [20] M. Lassas and G. Uhlmann, *On determining a Riemannian manifold from the Dirichlet-to-Neumann map*, Ann. Sci. École Norm. Sup. (4) 34 (2001), 771–787.
- [21] M. Levitin, L. Parnowski, and I. Polterovich, *Isospectral domains with mixed boundary conditions*, J. Phys. A: Math. Gen. 39 (2006), 2073–2082.
- [22] M. Levitin, L. Parnowski, I. Polterovich, and D. A. Sher, *Sloshing, Steklov and corners: Asymptotics of Steklov eigenvalues for curvilinear polygons*, Proc. London Math. Soc. 125 (2022), 359–436.
- [23] M. Levitin, L. Parnowski, I. Polterovich, and D. A. Sher, *Steklov problems on planar domains with cusps*, preprint, 2023.
- [24] W. McLean, *Strongly Elliptic Systems and Boundary Integral Equations*, Cambridge Univ. Press, 2000.
- [25] I. Polterovich, *Heat invariants of Riemannian manifolds*, Israel J. Math. 119 (2000), 239–252.
- [26] R. Plastock, *Homeomorphisms between Banach spaces*, Trans. Amer. Math. Soc. 200 (1974), 169–183.
- [27] I. Polterovich and D. A. Sher, *Heat invariants of the Steklov problem*, J. Geom. Anal. 25 (2015), 924–950.
- [28] H. M. Stark, *L-functions at $s = 1$. II. Artin L-functions with rational characters*, Adv. Math. 17 (1975), 60–92.
- [29] T. Sunada, *Riemannian coverings and isospectral manifolds*, Ann. of Math. (2) 121 (1985), 169–186.
- [30] J. Sylvester and G. Uhlmann, *A global uniqueness theorem for an inverse boundary value problem*, Ann. of Math. (2) 125 (1987), 153–169.
- [31] M. van den Berg and S. Srisatkunarakaj, *Heat equation for a region in $\mathbb{R}^2$ with a polygonal boundary*, J. London Math. Soc. 37 (1988), 119–127.

BLUECHIPS RESEARCH, WHARTON SCHOOL, UNIVERSITY OF PENNSYLVANIA, PHILADELPHIA, PA 19104
 Email address: rkulati@wharton.upenn.edu